

Sarah Lamik

Paris, France | sarahlamik9@gmail.com | +33 6 95 39 07 28 | [Projects portfolio](#)

Available for full-time positions from November 2026

ML engineer with research experience in designing and running independent experiments across reinforcement learning, graph neural networks, self-supervised learning and JAX-based implicit deep learning, with a focus on ablation-driven analysis of model behavior.

Education

CentraleSupélec, Université Paris-Saclay – ranked 2nd best engineering school in France Sept. 2022 – Apr. 2026
Engineering Degree, specialization in Artificial Intelligence - GPA: 4.09 / 4.33 (S1), 4.26/4.33 (S2)

- Relevant coursework: Statistical Learning, Convex Optimization, Reinforcement Learning (MDPs, bandits, policy gradient), Graph Machine Learning, Probabilistic Modeling, SAT Solving and Combinatorial Reasoning, Game Theory, Signal Processing, Cloud Computing.

Lycée Hoche – CPGE PC*

Sept. 2020 – Jun. 2022

- Intensive two-year selective program in Mathematics and Physics.
- Advanced Mathematics, Algorithms, Complexity Theory and Theoretical Physics.

Research Experience

Research Intern – RTE & Mines Paris (Ongoing)

May 2026 – Oct. 2026

Graph Neural Networks and Implicit Deep Learning

- Co-authoring a **scientific publication** on the intersection of implicit deep learning and GNN prediction
- Contributing via PR-based workflow to EnerGNN, an **open-source** GNN library (in progress)
- Building a Deep Equilibrium (DEQ) coupler from scratch in JAX, modeling power-grid state as a heterogeneous hypergraph.
- Implementing implicit differentiation and sparsity-exploiting linear solvers for fixed-point inference.
- *Current state:* forward and backward pass implemented.
- *Ongoing work:* scaling training to graphs of 50K+ nodes using distributed **data parallelism (DDP)** and **model parallelism**.

Academic Research Projects

Explainable Resource Allocation and Data Disclosure

2026 - [Research report](#)

Paris-Saclay University, CentraleSupélec, Laboratory of Mathematics and Computer Science (MICS)

- Studied the privacy risks induced by explanations generated in SAT-based resource allocation systems.
- Formalized contrastive explanations using Minimum Unsatisfiable Sets (MUS).
- Designed quantitative metrics for measuring information disclosure.
- Investigated the trade-off between explainability and privacy under rank-k envy-freeness constraints.
- Tools: PySAT, SAT solving, MUS extraction.

Autonomous Driving with Reinforcement Learning

2026 - [GitHub](#) & [Research report](#)

- Implemented and compared DQN, Double-DQN and Prioritized Experience Replay.
- Performed behavioral and performance analyses of trained agents.
- Evaluated the impact of algorithmic choices through ablation studies.

Graph Neural Networks: GCN vs GAT Ablation Study

2026 - [GitHub](#)

- Investigated the architectural trade-offs between GCN and GAT for node classification and link prediction on large-scale graphs (git_web_ml: 37K nodes, 290K edges; WikiOGB)
- Implemented hard negative sampling to address the false negative issue in link prediction evaluation.
- Best model (3-layer GCN + MLP decoder) achieved 93.3% link prediction accuracy; best classification model (2-layer GAT) reached 78.3% accuracy.
- Tools: PyTorch Geometric, scikit-learn, Node2Vec baseline.

Explainability in Linear Optimization

Reference letter by Vincent Mousseau (MICS, CentraleSupélec)

2026 - [GitHub](#)

- Developed algorithms for generating (1-1), (1-m), (m-1) and mixed explanations in optimization problems.
- Conducted large-scale experiments on transportation and healthcare datasets.
- Studied the existence conditions of optimization-based explanations.
- Tools: Gurobi, scikit-learn, NetworkX.

Industry AI Experience

European Central Bank - [GitLab link](#)

Jan. 2025 – Jun. 2025

Risk Management Department - Received an official letter of recommendation

- Designed a **RAG assistant**: tailored chunking strategy, benchmarked embedding retrieval vs. TF-IDF baseline.
- Tool used by ~20 people to retrieve information in 13 internal reports of ~150 pages.
- Developed anomaly detection tools for financial risk monitoring.

Audacia – Venture Capital

Jul. 2024 – Dec. 2024

- Conducted technical due diligence on AI startups, evaluated scientific and technological feasibility of AI-based industrial solutions.

Technical Skills

Deep Learning & ML: PyTorch, PyTorch Geometric, JAX, TensorFlow, scikit-learn, NumPy, pandas

Optimization & Symbolic AI: Gurobi, PySAT, Constraint Programming, SAT Solving

Infrastructure : Git, Linux, SQL, AWS, CI/CD

Web/App: React, Node.js, Streamlit

Languages

French (native), English (fluent, C1), German (intermediate, B1)